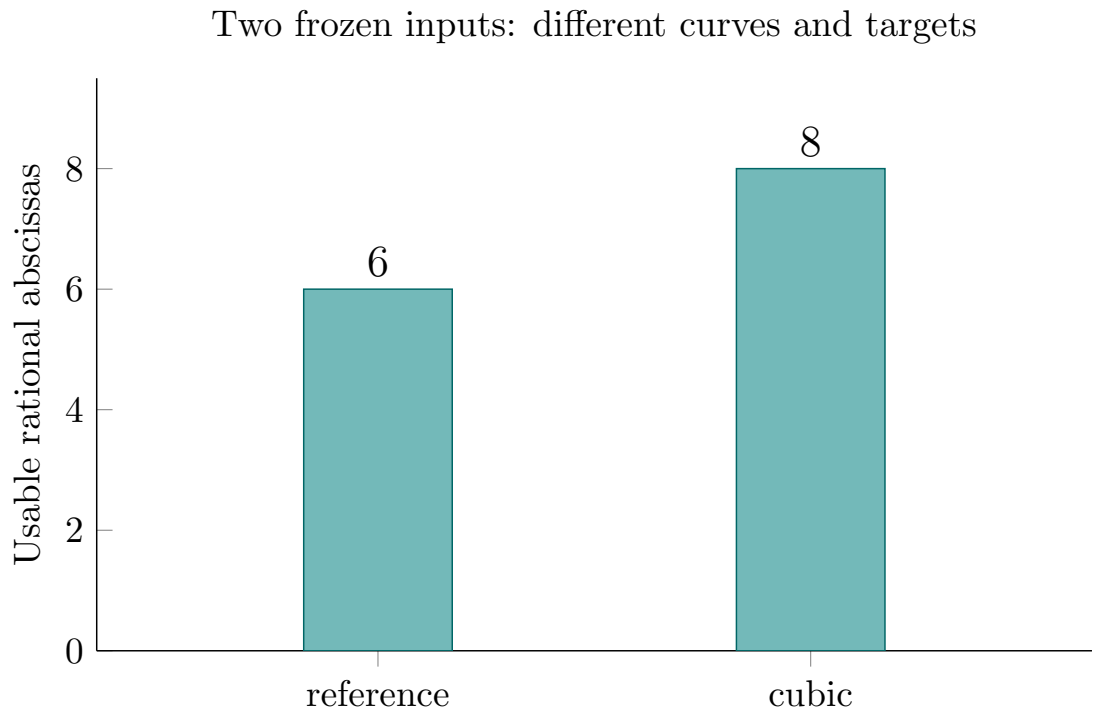
Saved cover inputs: occupancy and representation

November 13, 2026. Unsigned native audit [pcc:unverified].

The two completed round5 records each have16 distinct image abscissas, but only6 reference and8 cubic abscissas pass the nonzero-square filter. The cubic count matches its saved field; the reference count was absent. Both target abscissas are rational on their respective curves. These are different curves and planted targets, not a general cubic-map density advantage.



Exact check. The frozen Rust validator `bf27909df` pins both input digests, checks the31-bit prime and curve nonsingularity, primitive16th root, coset distinctness, images and all32 point/residue calculations. Neither case has a zero-y abscissa. Full point records are retained. It does not replay F4, its basis or planted solution, prove subgroup order, recover a scalar or measure useful-relation yield. Statement `PCC-20261113-79248f` pins the unsigned data artifact; PCC reports not_covered.

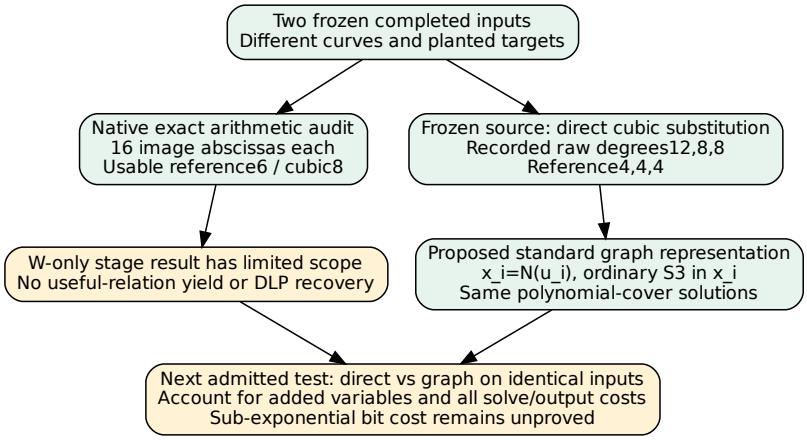
Original gate retained. The reported W-only cubic cell remains C: 245930 versus167726 columns. Its same-seed retry alone completed after the shared-pod kill. Quart/generic3 remain censored. The producer corrected the late erroneous hard-stop text and preserved it. Different curve/target draws, only one planted target per arm and mixed concurrency limit total-cost interpretation. This audit does not change the frozen finite W gate.

Encoding observation. Frozen source `3466f3be9e` substitutes cubic $N(u)$ directly into S3; raw input degrees12,8,8 versus4,4,4 for the reference. The recorded last productive degree is7. Explicit image-coordinate variables are a distinct standard representation to test.

Graph correspondence and the proposed next test

Introduce $x_i = N(u_i)$ and use ordinary S3 in the new image coordinates. For this polynomial cover there are no denominator guards. The proposed m4 layout has22 variables/23 equations and maximum input degree4. These are unimplemented-builder predictions. Added variables may outweigh any benefit; lower input degree does not imply less work or a sub-exponential bound.

Formal map. `CoverGraph.lean`, frozen source `75e4b203e`, defines original solutions, lifted solutions and the actual lift/project maps for arbitrary map ϕ and residual predicate. The theorems projectLift and liftProject establish both inverse identities. This proves generic pointwise solution correspondence, not an actual polynomial builder, degree bound or solver complexity. It is standard graph substitution, with no novelty claim.



Checks and failures. Native16 tests pass after adding the omitted pinned hex test dependency; failed test001 retained. Graph builds001/002 fail on opaque subtype compilation/core-only Type* syntax;003 uses exposed aliases and explicit universes, completes3 jobs and reports no axioms for either identity. Source audit and fresh import-inclusive replay have attributable receipts. The premature failed-source commit remains a failed stage.

Next admitted experiment. Compare direct and graph systems on exactly the same literal curve, coset, map and target. Validate both projection directions and known-answer witnesses, then count actual sparse work and all solve/output costs. Measure positive and naturally negative cases separately. Record usable base sizes and setup rejection costs; do not assume relation yield from the observed6/8 pair. Preserve memory failures and isolate resources for time comparisons. The draft protocol remains unexecuted.

Sources and resources. Original protocol and amendments, launch scripts, saved rows, progress log and binary digest; frozen map, summation and generator code; prior auxiliary-cover review. KB MCP/wrapper unavailable, bounded GitHub fallback. Native correctness only: no new target, solver, timing/scaling run or paid job. Monthly isolab budget remains exhausted. No benchmark, leaderboard or ecdb result. A sub-exponential public-input finder and its bit cost remain open. Editable source, data, DOT/SVG and exact receipts accompany this report.